Algoritmi Avansați 2026
c-7
Randomized Data Structures: Skip Lists; Bloom Filters

Lect. Dr. Ștefan Popescu

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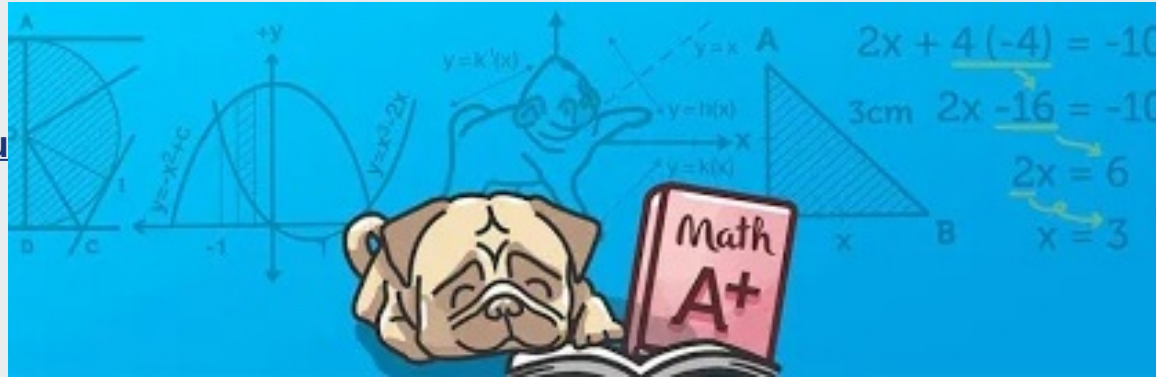

Grup Teams:

Randomized Data Structures

- O privire pe scurt, “din avion” asupra Skip lists.

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Skip lists are a probabilistic data structure that seem likely to supplant balanced trees as the implementation method of choice for many applications. Skip list algorithms have the same asymptotic expected time bounds as balanced trees and are simpler, faster and use less space.

— William Pugh, Concurrent Maintenance of Skip Lists (1989)

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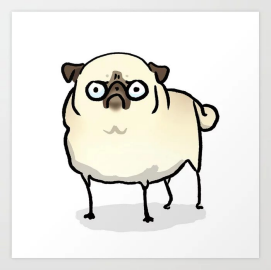
A: NU!



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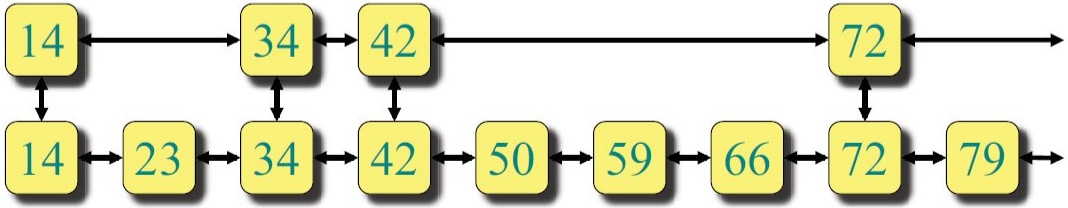
A: O(log n)

Skip Lists

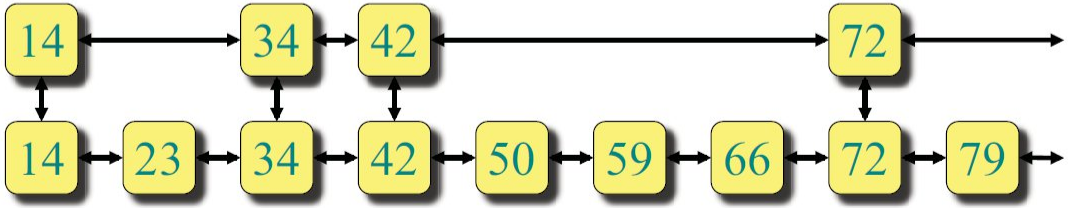
Pentru o cautare mai eficienta, vom retine 2 liste, dupa modelul urmator:



Skip Lists

SEARCH(x):

- Walk right in top linked list (L1) until going right would go too far
- Walk down to bottom linked list (L2)
- Walk right in L2 until element found (or not)



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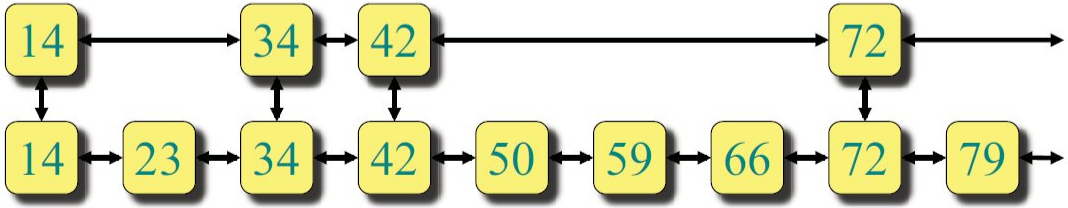

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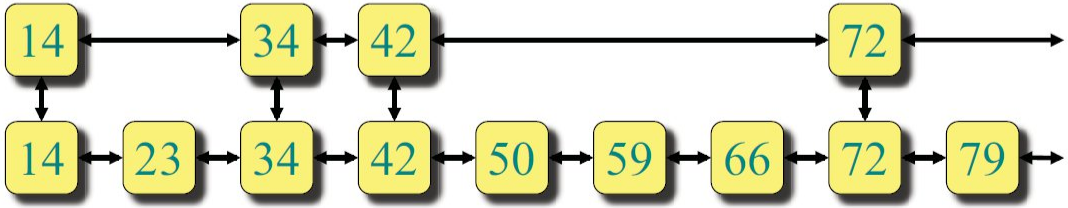

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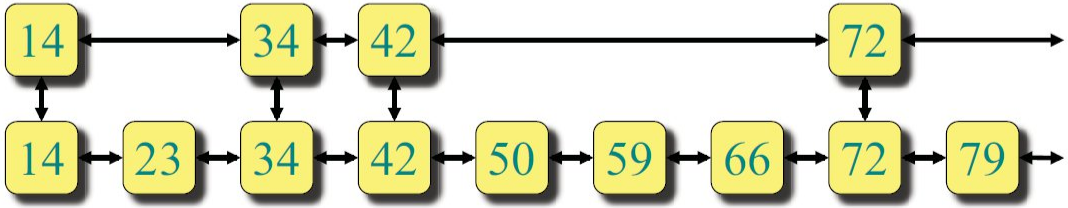

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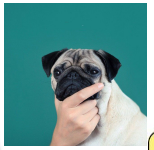
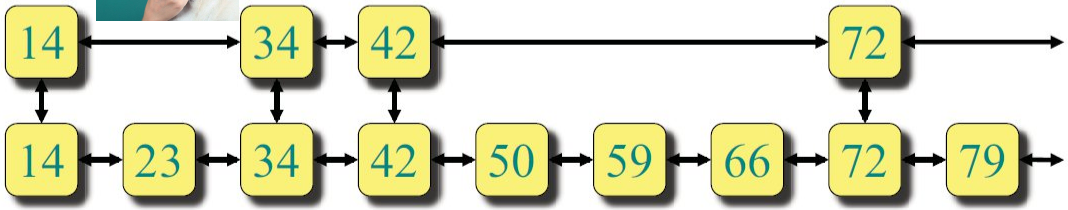
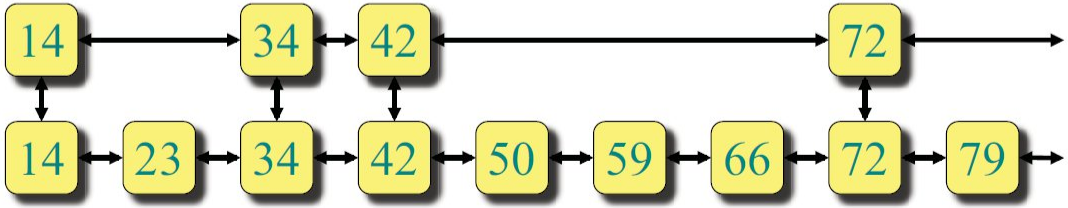




Skip Lists: Numarul de elemente per nivel

Costul unei căutări în listă este aprox. $|L1| + \frac{|L2|}{|L1|}$



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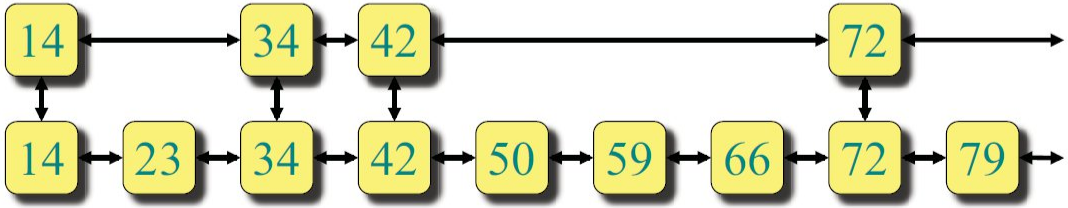
Costul unei căutări în listă este aprox $|L1| + \frac{|L2|}{|L1|}$

Relatia de mai sus este minimizata atunci cand $|L1|^2=|L2|=n$; deci

$|L1|=sqrt(n)$



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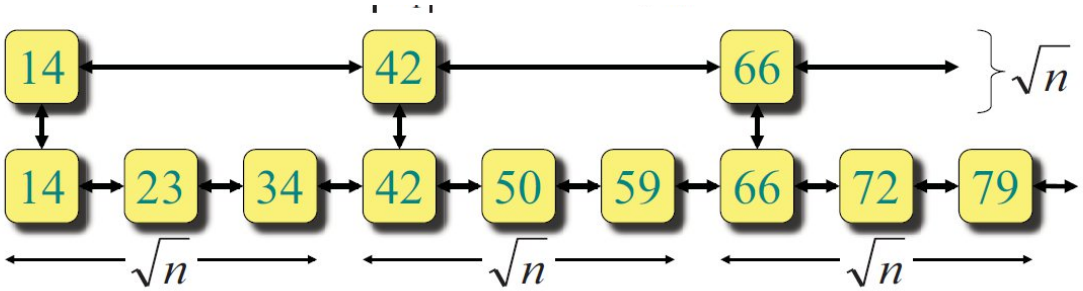
$|L1| = \sqrt{n}$; $|L2| = n$; *Avem costul total de cautare pe 2 nivele*:

$$|L1| + \frac{|L2|}{|L1|} = 2\sqrt{n}$$



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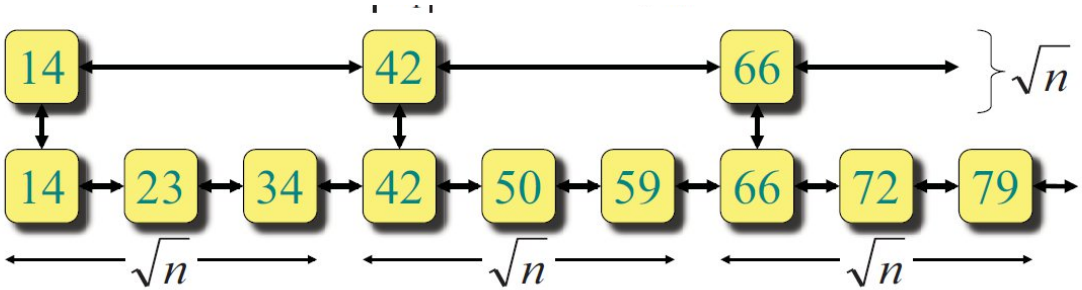
Dar pentru 3 nivele?
Dar 4 nivele?
Dar *k* nivele?



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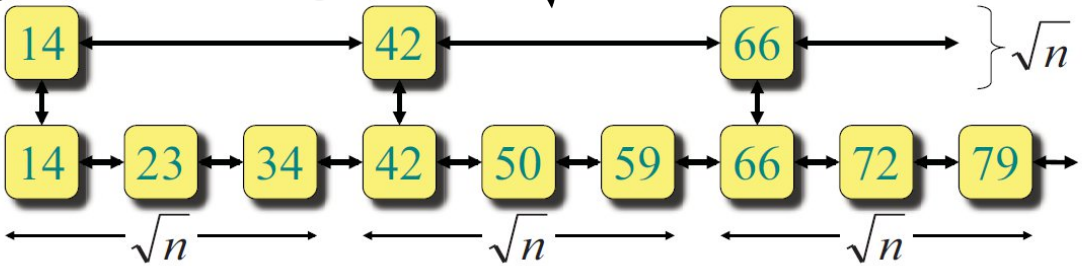
$2\ nivele: 2\sqrt{n}$

$3\ nivele: 3\sqrt[3]{n}$

$k\ nivele: k\sqrt[k]{n}$

$\lg(k)\ nivele: lg(.k).\sqrt[lg(k)]{n}$



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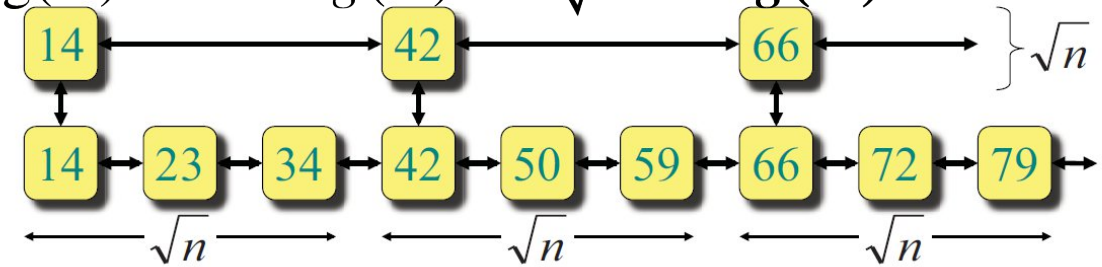

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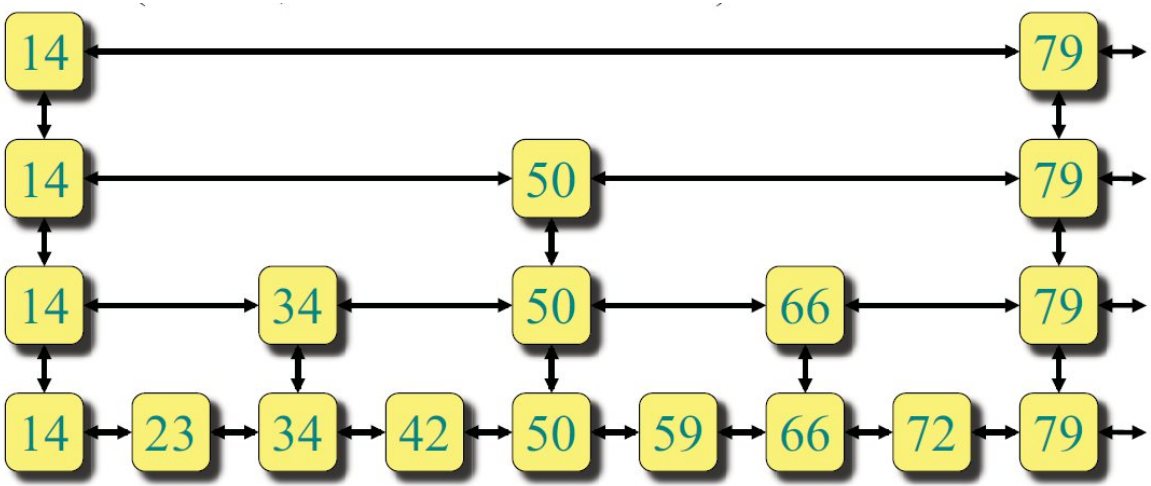
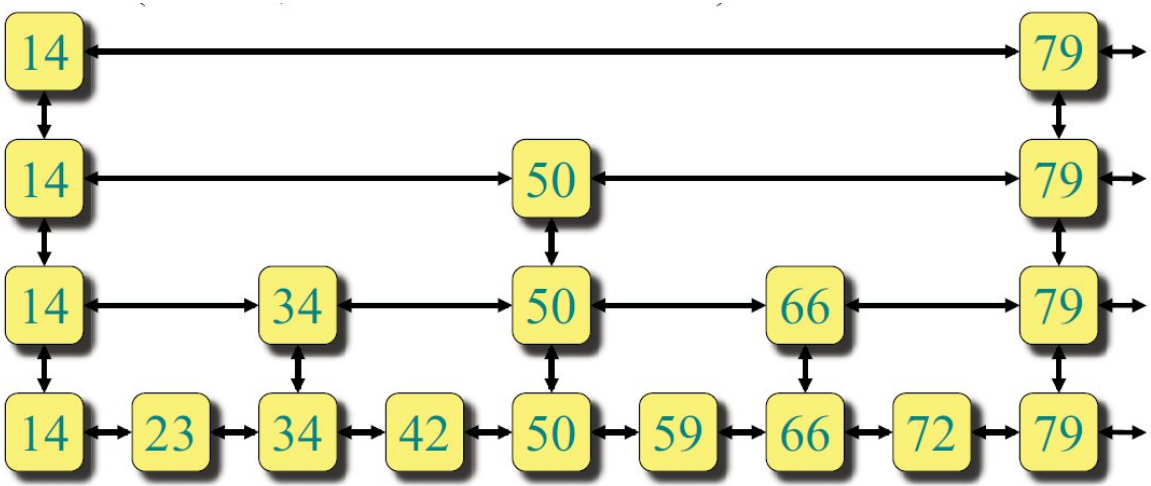


Cu ce seamănă oare?

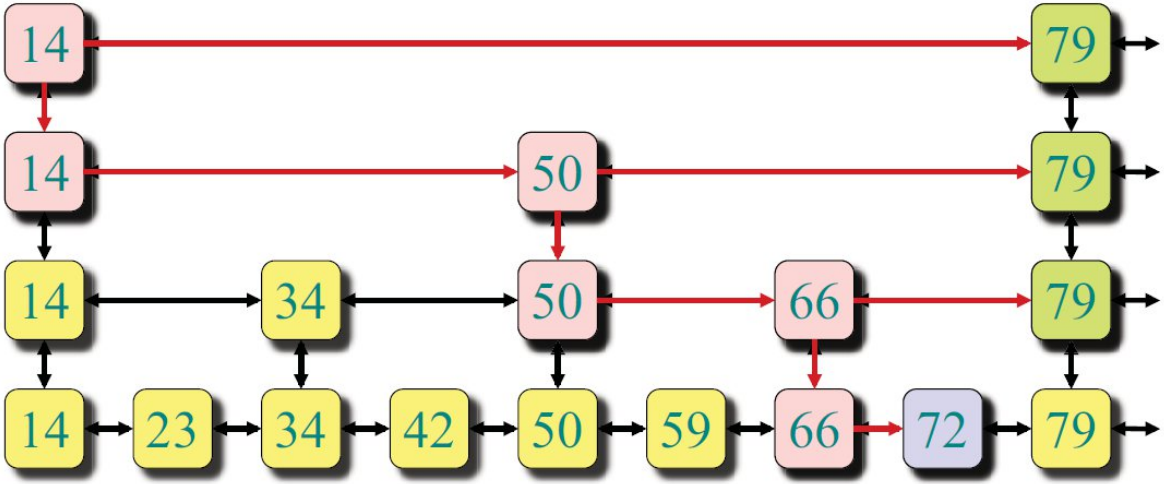
Skip Lists: Numarul de elemente per nivel



Cu ce seamana oare?

Un arbore!

Skip Lists: Search (X)



Search(72)

Skip Lists: Insert (X)

Pentru a insera un nou element X in lista:

- **ii cautam pozitia in nivelul inferior (search(x))**
- **il inseram pe nivelul inferior**

Skip Lists: Insert (X)

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OBSERVATIE: Nivelul inferior va contine intoteauna toate elementele

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A: Dau cu banul! Daca pica pajura, inserez pe inca un nivel, altfel ma opresc!

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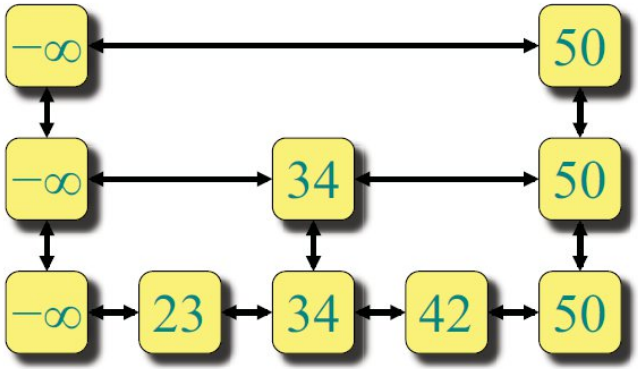
Consecinta: ½ dintre elemente vor fi doar pe nivelul 0. ¼ dintre elemente vor fi doar pe nivelele 0 si 1. ⅛ vor fi doar pe nivelele 0, 1 si 2, etc...

Skip Lists: Exercitiu

EXERCISE: Try building a skip list from scratch by repeated insertion using a real coin

Small change:

- Add special $-\infty$ value to *every* list $\Rightarrow$ can search with the same algorithm



Skip Lists: Delete (x)

Se cauta elementul x in lista (se gaseste pe cel mai de sus nivel).

Se sterge elemntul de pe toate nivelele!

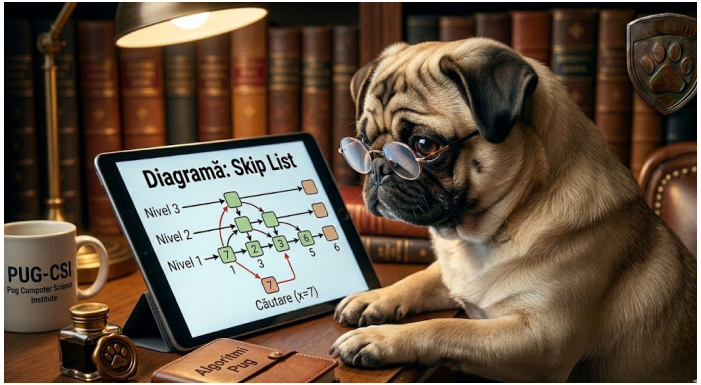
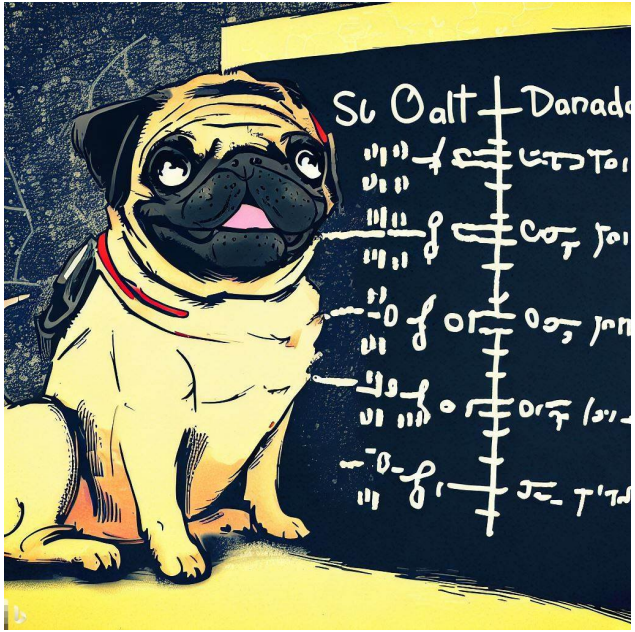
Skip Lists: How good are they?

[Whiteboard]

[Mit Course on Skip Lists]

[Lecture Notes]

[Lecture Slides]



Bonus: Bloom Filters